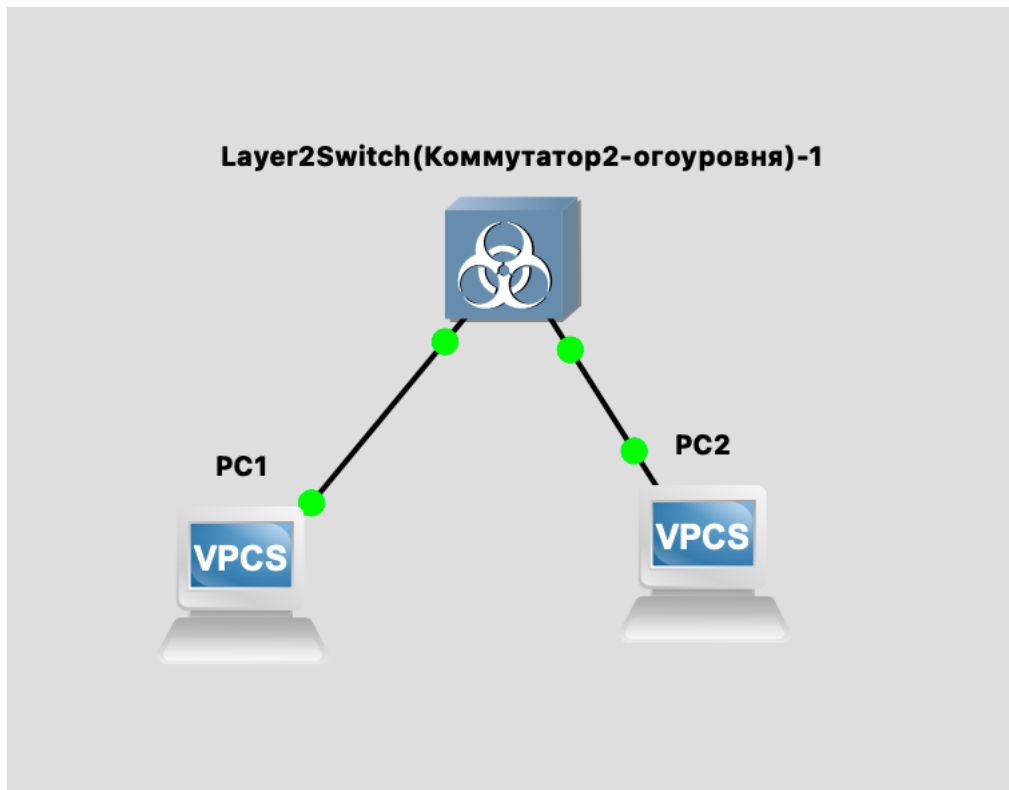




Topology 1: switch and two pcs

```
PC1> ip 192.168.1.1 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0
```

```
PC2> ip 192.168.1.2 255.255.255.0
Checking for duplicate address...
PC2 : 192.168.1.2 255.255.255.0
```

PC1's config:

```
set pcname PC1
ip 192.168.1.1 255.255.255.0 24
```

PC2's config:

```
set pcname PC2
ip 192.168.1.2 255.255.255.0 24
```

ping PC1 to PC2:

PC1> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=12.632 ms

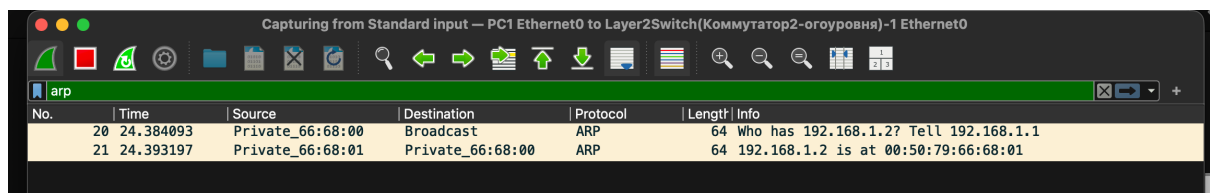
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=1.667 ms

84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=7.512 ms

84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=6.644 ms

^C

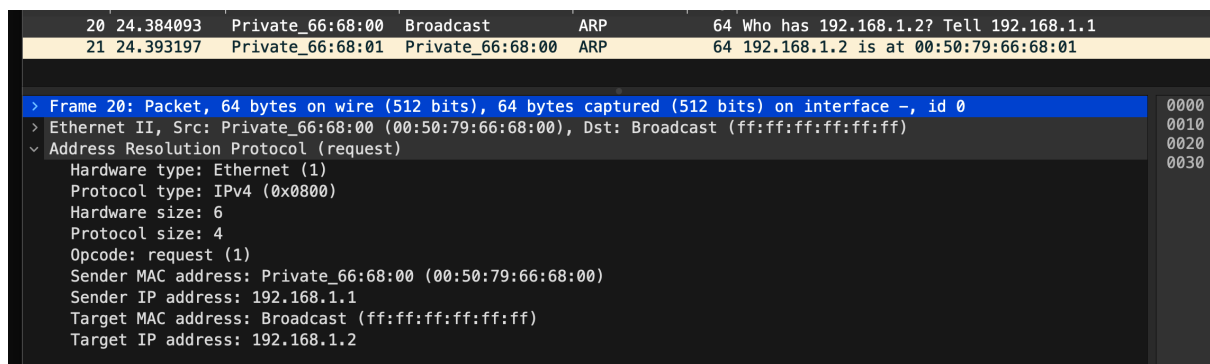
PC1>



The image shows a Wireshark packet capture window. The title bar reads "Capturing from Standard input - PC1 Ethernet0 to Layer2Switch(Коммутатор2-огоуровня)-1 Ethernet0". The filter bar contains "arp". The packet list shows two packets:

No.	Time	Source	Destination	Protocol	Length	Info
20	24.384093	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.2? Tell 192.168.1.1
21	24.393197	Private_66:68:01	Private_66:68:00	ARP	64	192.168.1.2 is at 00:50:79:66:68:01

Capture link 1: from PC1 to switch



The image shows the packet details view for the first packet (No. 20). The packet is an ARP request. The details are as follows:

Frame 20: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0	0000
Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)	0010
Address Resolution Protocol (request)	0020
Hardware type: Ethernet (1)	0030
Protocol type: IPv4 (0x0800)	
Hardware size: 6	
Protocol size: 4	
Opcode: request (1)	
Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)	
Sender IP address: 192.168.1.1	
Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)	
Target IP address: 192.168.1.2	

ARP request analysis:

Hardware type: Ethernet

Protocol type: IPv4

Opcode: request

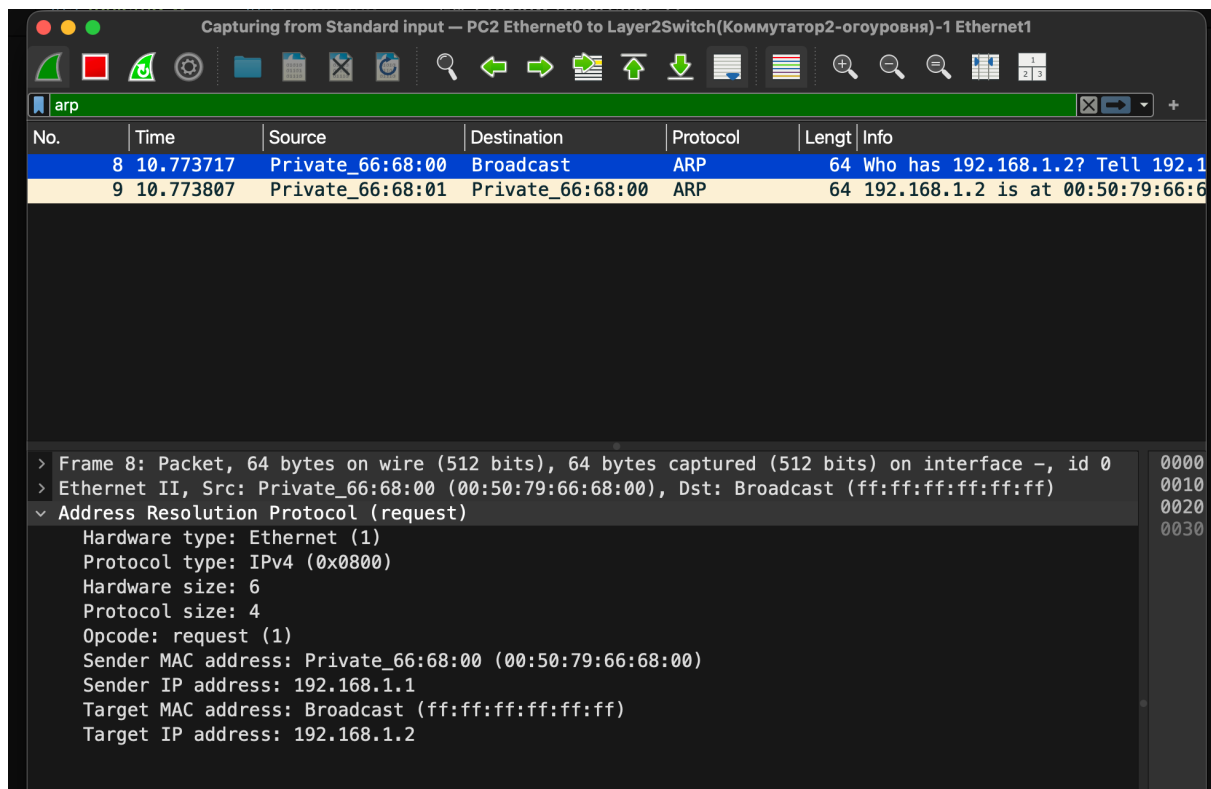
Sender MAC address: 00:50:79:66:68:00

Sender IP address: 192.168.1.1

Target MAC address: ff:ff:ff:ff:ff:ff (Broadcast)

Target IP address: 192.168.1.2

Conclusion: device 192.168.1.1 finds MAC-address device 192.168.1.2, broadcasting ARP request. Waiting for device 2 reply with own MAC.



Capture link 2: from switch to pc2

ARP request analysis:

Hardware type: Ethernet

Protocol type: IPv4

Opcode: request (1)

Sender MAC address: 00:50:79:66:68:00

Sender IP address: 192.168.1.1

Target MAC address: ff:ff:ff:ff:ff:ff (Broadcast)

Target IP address: 192.168.1.2

arp						
No.	Time	Source	Destination	Protocol	Length	Info
8	10.773717	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.2? Tell 192.1
9	10.773807	Private_66:68:01	Private_66:68:00	ARP	64	192.168.1.2 is at 00:50:79:66:6
> Frame 9: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0 > Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00) > Address Resolution Protocol (reply) Hardware type: Ethernet (1) Protocol type: IPv4 (0x0800) Hardware size: 6 Protocol size: 4 Opcode: reply (2) Sender MAC address: Private_66:68:01 (00:50:79:66:68:01) Sender IP address: 192.168.1.2 Target MAC address: Private_66:68:00 (00:50:79:66:68:00) Target IP address: 192.168.1.1						

ARP reply analysis:

Hardware type: Ethernet

Protocol type: IPv4

Opcode: reply (2)

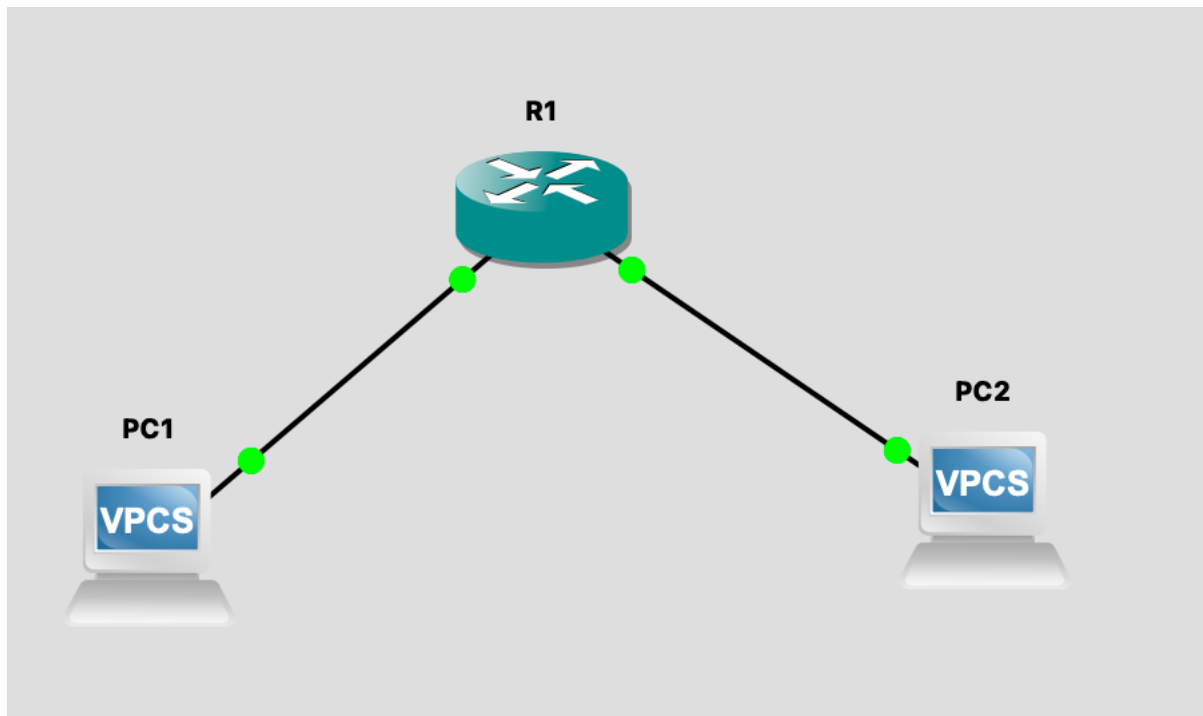
Sender MAC address: 00:50:79:66:68:01

Sender IP address: 192.168.1.2

Target MAC address: 00:50:79:66:68:00

Target IP address: 192.168.1.1

Conclusion: device 192.168.1.2 replies to ARP request with his MAC 00:50:79:66:68:00.



Topology 2: router and two pcs

```
PC1> set pcname PC3
```

```
PC3> ip 192.168.2.1 192.168.2.5
```

```
Checking for duplicate address...
```

```
PC3 : 192.168.2.1 255.255.255.0 gateway 192.168.2.5
```

```
PC2> set pcname PC4
```

```
PC4> ip 192.168.1.1 192.168.1.5
```

```
Checking for duplicate address...
```

```
PC4 : 192.168.1.1 255.255.255.0 gateway 192.168.1.5
```

```

R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.2.5 255.255.255.0
R1(config-if)#duplex auto
R1(config-if)#speed auto
R1(config-if)#no shutdown
R1(config-if)#!
R1(config-if)#exit
*Mar  1 00:10:18.559: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:10:19.559: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#interface FastEthernet1/0
R1(config-if)#ip address 192.168.1.5 255.255.255.0
R1(config-if)#duplex auto
R1(config-if)#speed auto
R1(config-if)#no shu
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:10:45.707: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar  1 00:10:46.707: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R1(config)#

```

```

R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.2.5 255.255.255.0
R1(config-if)#duplex auto
R1(config-if)#speed auto
R1(config-if)#no shutdown
R1(config-if)#!
R1(config-if)#exit
*Mar  1 00:10:18.559: %LINK-3-UPDOWN: Interface
FastEthernet0/0, changed state to up
*Mar  1 00:10:19.559: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#interface FastEthernet1/0
R1(config-if)#ip address 192.168.1.5 255.255.255.0
R1(config-if)#duplex auto
R1(config-if)#speed auto
R1(config-if)#no shu

```

```
R1(config-if)#no shutdown
```

```
R1(config-if)#exit
```

```
R1(config)#
```

```
*Mar  1 00:10:45.707: %LINK-3-UPDOWN: Interface  
FastEthernet1/0, changed state to up
```

```
*Mar  1 00:10:46.707: %LINEPROTO-5-UPDOWN: Line protocol on  
Interface FastEthernet1/0, changed state to up
```

```
R1(config)#
```

```
[PC3> ping 192.168.1.1
```

```
192.168.1.1 icmp_seq=1 timeout
```

```
192.168.1.1 icmp_seq=2 timeout
```

```
84 bytes from 192.168.1.1 icmp_seq=3 ttl=63 time=14.850 ms
```

```
84 bytes from 192.168.1.1 icmp_seq=4 ttl=63 time=14.980 ms
```

```
84 bytes from 192.168.1.1 icmp_seq=5 ttl=63 time=15.126 ms
```

```
PC3> 
```

Ping PC3 to PC4

Capturing from Standard input - PC1 Ethernet0 to R1 FastEthernet0/0

No.	Time	Source	Destination	Protocol	Length	Info
2	9.429875	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.2.5? Tell 192.168.2.1
3	9.436223	cc:01:0d:87:00:00	Private_66:68:00	ARP	60	192.168.2.5 is at cc:01:0d:87:00:00
4	9.437332	192.168.2.1	192.168.1.1	ICMP	98	Echo (ping) request id=0x48f1, seq=1/256, ttl=64 (reply in 5)
5	9.466439	192.168.1.1	192.168.2.1	ICMP	98	Echo (ping) reply id=0x48f1, seq=1/256, ttl=63 (request in 4)
7	10.467055	192.168.2.1	192.168.1.1	ICMP	98	Echo (ping) request id=0x49f1, seq=2/512, ttl=64 (reply in 8)
8	10.482554	192.168.1.1	192.168.2.1	ICMP	98	Echo (ping) reply id=0x49f1, seq=2/512, ttl=63 (request in 7)
9	11.483038	192.168.2.1	192.168.1.1	ICMP	98	Echo (ping) request id=0x4af1, seq=3/768, ttl=64 (reply in 10)
10	11.498448	192.168.1.1	192.168.2.1	ICMP	98	Echo (ping) reply id=0x4af1, seq=3/768, ttl=63 (request in 9)
11	12.499045	192.168.2.1	192.168.1.1	ICMP	98	Echo (ping) request id=0x4bf1, seq=4/1024, ttl=64 (reply in 12)
12	12.514600	192.168.1.1	192.168.2.1	ICMP	98	Echo (ping) reply id=0x4bf1, seq=4/1024, ttl=63 (request in 11)

> Frame 2: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Address Resolution Protocol (request)

capture link from pc3 to router

ARP request:

```
> Frame 2: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
    Sender IP address: 192.168.2.1
    Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
    Target IP address: 192.168.2.5
```

ARP reply:

```
> Frame 3: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
> Ethernet II, Src: cc:01:0d:87:00:00 (cc:01:0d:87:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)
  Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: cc:01:0d:87:00:00 (cc:01:0d:87:00:00)
    Sender IP address: 192.168.2.5
    Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
    Target IP address: 192.168.2.1
```

ICMP request:

```
> Frame 4: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: cc:01:0d:87:00:00 (cc:01:0d:87:00:00)
> Internet Protocol Version 4, Src: 192.168.2.1, Dst: 192.168.1.1
  Internet Control Message Protocol
    Type: Echo (ping) request (8)
    Code: 0
    Checksum: 0xd719 [correct]
    [Checksum Status: Good]
    Identifier (BE): 18673 (0x48f1)
    Identifier (LE): 61768 (0xf148)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Response frame: 5]
  Data (56 bytes)
    Data: 08090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f
    [Length: 56]
```

ICMP reply:

```
> Frame 5: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
> Ethernet II, Src: cc:01:0d:87:00:00 (cc:01:0d:87:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)
> Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.2.1
  Internet Control Message Protocol
    Type: Echo (ping) reply (0)
    Code: 0
    Checksum: 0xdf19 [correct]
    [Checksum Status: Good]
    Identifier (BE): 18673 (0x48f1)
    Identifier (LE): 61768 (0xf148)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Request frame: 4]
    [Response time: 29.107 ms]
  Data (56 bytes)
    Data: 08090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f
    [Length: 56]
```


arp or icmp						
No.	Time	Source	Destination	Protocol	Length	Info
3	14.628617	192.168.2.1	192.168.1.1	ICMP	98	Echo (ping) request id=0x83f2, seq=1/256, ttl=63 (reply in 6)
4	14.628723	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.5? Tell 192.168.1.1
5	14.638692	cc:01:0d:87:00:10	Private_66:68:01	ARP	60	192.168.1.5 is at cc:01:0d:87:00:10
6	14.639346	192.168.1.1	192.168.2.1	ICMP	98	Echo (ping) reply id=0x83f2, seq=1/256, ttl=64 (request in 3)
7	15.654894	192.168.2.1	192.168.1.1	ICMP	98	Echo (ping) request id=0x84f2, seq=2/512, ttl=63 (reply in 8)
8	15.655012	192.168.1.1	192.168.2.1	ICMP	98	Echo (ping) reply id=0x84f2, seq=2/512, ttl=64 (request in 7)
9	16.670897	192.168.2.1	192.168.1.1	ICMP	98	Echo (ping) request id=0x85f2, seq=3/768, ttl=63 (reply in 10)
10	16.671000	192.168.1.1	192.168.2.1	ICMP	98	Echo (ping) reply id=0x85f2, seq=3/768, ttl=64 (request in 9)

Capture link from router to pc4

Icmp req:

```
> Frame 3: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
> Ethernet II, Src: cc:01:0d:87:00:10 (cc:01:0d:87:00:10), Dst: Private_66:68:01 (00:50:79:66:68:01)
> Internet Protocol Version 4, Src: 192.168.2.1, Dst: 192.168.1.1
> Internet Control Message Protocol
  Type: Echo (ping) request (8)
  Code: 0
  Checksum: 0x9c18 [correct]
  [Checksum Status: Good]
  Identifier (BE): 33778 (0x83f2)
  Identifier (LE): 62083 (0xf283)
  Sequence Number (BE): 1 (0x0001)
  Sequence Number (LE): 256 (0x0100)
  [Response frame: 6]
> Data (56 bytes)
```

Arp req:

```
> Frame 4: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
  Sender IP address: 192.168.1.1
  Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
  Target IP address: 192.168.1.5
```

Arp rep:

```
> Frame 5: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
> Ethernet II, Src: cc:01:0d:87:00:10 (cc:01:0d:87:00:10), Dst: Private_66:68:01 (00:50:79:66:68:01)
> Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: cc:01:0d:87:00:10 (cc:01:0d:87:00:10)
  Sender IP address: 192.168.1.5
  Target MAC address: Private_66:68:01 (00:50:79:66:68:01)
  Target IP address: 192.168.1.1
```

Icmp rep:

```
> Frame 6: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: cc:01:0d:87:00:10 (cc:01:0d:87:00:10)
> Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.2.1
< Internet Control Message Protocol
  Type: Echo (ping) reply (0)
  Code: 0
  Checksum: 0xa418 [correct]
  [Checksum Status: Good]
  Identifier (BE): 33778 (0x83f2)
  Identifier (LE): 62083 (0xf283)
  Sequence Number (BE): 1 (0x0001)
  Sequence Number (LE): 256 (0x0100)
  [Request frame: 3]
  [Response time: 10.729 ms]
> Data (56 bytes)
```